MESA COHORT C4

AI and its Applications — Section 1 Final Examination

PROBLEM STATEMENT BOOKLET

Instructions to Students

1. Read your assigned problem statement carefully.
2. Your solution must use live external data wherever possible. The system must work with some data at the time of submission.
3. All AI tasks must use Gemini Flash. Do not use Claude API.
4. Deploy your solution on Vercel. Submit a working public URL along with your GitHub repository link and the Components Map.

Medium Difficulty

Industry Event Scout

Business Development • Professional Networking

Vikram is an independent management consultant whose practice generates forty percent of its new business through in-person events - conferences, roundtables, CXO summits, and startup ecosystem meetups. When he meets a potential client face-to-face, his conversion rate is three times higher than any cold outreach channel. The problem is not attending events - it is finding them in the first place. Relevant events in his space are announced across EventBrite, iSpirit, conference-specific websites, and a dozen industry newsletters. There is no single source.

He currently spends forty-five minutes every Monday morning manually checking these sources, bookmarking events, and adding deadlines to his calendar. He still misses things. Last month, he found out about a CIO roundtable in Bengaluru the day after the registration deadline - it was exactly the right room for three of his active prospects. He missed it because the organiser only posted on LinkedIn and he didn't see it in time.

Vikram wants a Monday morning email that consolidates everything relevant happening in his professional domains across the next thirty days - regardless of where those events were announced. He doesn't want to manually cross-check multiple platforms every week. He wants the system to aggregate from every source where such events appear, filter for his domains and target cities, and rank results by how closely each event matches his stated focus areas. The digest should be scannable in under five minutes: a clean list with enough context per event to make a quick decision on whether to register. His target categories and cities should be configurable without touching code, since his practice focus shifts across the year. Any event with a registration deadline fewer than seven days away must be clearly flagged - he has already learned that lesson the hard way.

> Build and deploy an AI-powered weekly event intelligence digest for Vikram, delivered every Monday morning before he starts his week.